AI/ML Open Source Ecosystem 2025

Power BI Dashboard Report — Comprehensive Overview of 1,200 Repositories Across 10 Categories

Report Type: Analytical Dashboard Documentation

Pages Covered: Executive · Popularity & Community · Technology · Trends

Data Source: AI/ML Open Source Repository Dataset (1,200 records)

Tooling: Microsoft Power BI Desktop · DAX · Deneb Custom Visuals

Contents

1. Project Background

The AI/ML Open Source Ecosystem 2025 dashboard was built to give a single, interactive view of the health, popularity, and technical composition of the open-source AI and machine-learning landscape. Open-source tooling underpins almost every stage of the modern AI stack — from data engineering and model training to inference and deployment — yet that ecosystem is fragmented across thousands of repositories, frameworks, languages, and licensing models. This project consolidates a representative sample of that ecosystem into a single Power BI semantic model so that stakeholders (engineering leadership, developer-relations teams, OSS program offices, and investors) can compare categories, frameworks, and growth trends side by side instead of researching each repository individually.

The report was designed as a four-page interactive dashboard — Executive, Popularity & Community, Technology, and Trends — that lets a user drill from a high-level portfolio summary down to repository-level detail using slicers, buttons for page navigation, and cross-filtering between visuals on every page.

Objectives

- Provide a single source of truth for repository counts, stars, and engagement across 10 AI/ML categories.
- Surface the frameworks, languages, and licenses that dominate the current open-source AI stack.
- Track repository growth, age, and download trends to highlight which projects are accelerating.
- Support data-driven decisions on which categories or frameworks to invest engineering or sponsorship effort in.

2. Dataset Description

The semantic model is built around one fact table, **Repositories** (1,200 rows, 54 columns), describing one AI/ML open-source repository per row, along with two supporting dimension tables and a pre-computed correlation matrix used for the heatmap visual on Page 2. Organization names in the sample (e.g. indie-dev-*, startup-*, plus recognizable ecosystem players) represent a broad simulated cross-section of maintainers, from individual developers to companies, spanning 226 distinct organizations and 75 subcategories.

Tables in the Model

Table	Rows	Role	Description
Repositories	1,200	Fact table	One row per repository: identity, popularity metrics, engineering-health signals, tech stack, and lifecycle fields.
Dim_ai_ml_monthly_trends	3,600	Dimension / trend	Monthly time series per repository — cumulative stars/forks, downloads, issues, PRs, commits, and star growth rate.
Dim_ai_ml_tech_stack	4,860	Dimension / bridge	Repository-to-dependency bridge table listing each dependency name and type, flagged as ML library or not.
CorrelationMatrix	49	Supporting	Pre-computed pairwise correlation coefficients between 7 key metrics, feeding the Page 2 heatmap.

Key Fields in the Repositories Fact Table

Group	Fields
Identity & taxonomy	repo_id, repo_name, organization, category, subcategory, popularity_tier
Popularity & engagement	stars, forks, watchers, open/closed issues, open/merged pull requests, contributors_count, total_commits, community_sentiment_score
Technology stack	primary_language, secondary_languages, framework_dependency, package_manager, license_type, hardware_requirement, cloud_integration, minimum_python_version
Lifecycle & activity	first_commit_date, last_commit_date, repo_age_days, is_actively_maintained, release_count, avg_issue_response_hours
Adoption	monthly_downloads, monthly_active_users, code_size_kb
Engineering-health flags	has_documentation, has_ci_cd, has_docker_support, has_unit_tests, has_changelog, has_arxiv_paper, has_live_demo, has_api, has_cli, has_gui, readme_quality
Derived / calculated columns	repo_health_score, Community Engagement Score, Star Range, Repository Age Group, Package Manager Display

3. Dashboard Design Process

The report follows a top-down, drill-friendly structure: a portfolio-level Executive summary, followed by three focused analytical pages, each reachable via on-canvas navigation buttons and each built at a 1920×1080 16:9 canvas size for full-screen presentation mode.

Data modeling

A single-fact star schema was built around the Repositories table, with a bridge table for tech-stack dependencies and a separate monthly-trends table for time-series analysis. Local date tables were auto-generated for each date column to support Year/Quarter/Month drill-downs on the Trends page.

Calculated columns

Several derived columns were added in DAX to make binning and display cleaner at the report layer: Star Range (bucketed star counts), Repository Age Group (six lifecycle buckets), Package Manager Display (friendlier labels via SWITCH), and Community Engagement Score (a weighted composite of stars, forks, watchers, and contributors).

Measures layer

A dedicated 'My Measures' measure table was created to keep all 16 DAX measures separate from the data tables, covering totals, averages, distinct counts, and a custom Growth Score composite metric (see Section 4).

Page layout

Each analytical page opens with a KPI card strip (4–6 cards) summarizing the page's focus metrics, followed by a grid of charts chosen to match the data type: bar/column charts for categorical comparisons, a treemap for part-to-whole share of a large category, donut charts for composition, a 100%-stacked bar for tier mix by category, an area chart for distribution shape, and a line chart for the repository-creation trend over time.

Custom visuals

Two Deneb (Vega-Lite) custom visuals were used for chart types Power BI doesn't support natively: a scatter plot (Stars vs. Health Score, and Repository Health vs. Repository Age) and matrix heatmaps (Metric Correlation, and Framework vs. Category). A tCard custom visual was also used for compact KPI presentation.

Navigation & interactivity

Bookmark-driven action buttons on the left rail let users jump between the four pages; slicers filter by category/tier; and every visual on a page cross-filters the others by default, so clicking a category bar on the Executive page filters the rest of that page's visuals.

Theming

A dark navy (#07182F) canvas with a consistent accent palette was applied report-wide via a custom Power BI theme, chosen to keep dense multi-chart pages readable and to make KPI card values pop against the background.

4. DAX Measures Created

All measures live in a dedicated **My Measures** table. The table below lists every measure exactly as defined in the model, together with a plain-language description of what it drives on the dashboard.

Measure	DAX Expression	Purpose on Dashboard
Total Repositories	`COUNTROWS(Repositories)`	Portfolio-size KPI card; base of Framework Usage and License Distribution donuts.
Total Stars	`SUM(Repositories[stars])`	Headline popularity KPI; used on Executive, Popularity, and Trends pages.
Average Health Score /100	`AVERAGE(Repositories[repo_health_score])`	Engineering-quality KPI (docs, CI/CD, tests, maintenance combined).
Average Sentiment (0-1)	`AVERAGE(Repositories[community_sentiment_score])`	Community-sentiment KPI on the Executive page.
Average Contributors	`AVERAGEX(FILTER(Repositories, NOT ISBLANK(Repositories[contributors_count])), Repositories[contributors_count])`	Average team size KPI; explicitly filters blanks so they don't skew the average.
Average Downloads	`AVERAGE(Repositories[monthly_downloads])`	Adoption KPI on the Executive page.
Active Repositories	`CALCULATE(COUNTROWS(Repositories), Repositories[is_actively_maintained]=TRUE())`	Filters to actively maintained projects for maintenance-health analysis.
Average Forks	`AVERAGE(Repositories[forks])`	Community KPI on Popularity and Trends pages.
Average Watchers	`AVERAGE(Repositories[watchers])`	Community KPI on Popularity and Trends pages.
Total Frameworks	`DISTINCTCOUNT(Repositories[framework_dependency])`	Technology-diversity KPI on the Technology page.
Total Languages	`DISTINCTCOUNT(Repositories[primary_language])`	Technology-diversity KPI on the Technology page.
Total Package Managers	`DISTINCTCOUNT(Repositories[package_manager])`	Technology-diversity KPI on the Technology page.
Total License Types	`DISTINCTCOUNT(Repositories[license_type])`	Licensing-diversity KPI on the Technology page.
Average Repository Health	`AVERAGE(Repositories[repo_health_score])`	Duplicate-purpose measure used specifically in the Technology-page multi-row cards.
Average Community Engagement	`AVERAGE(Repositories[Community Engagement Score])`	Averages the composite engagement calculated column for the Technology page.
Growth Score	`DIVIDE(SUM(Repositories[stars]),1000)*0.40 + DIVIDE(AVERAGE(Repositories[monthly_downloads]),1000)*0.30 + DIVIDE(AVERAGE(Repositories[monthly_active_users]),1000)*0.20 + DIVIDE(AVERAGE(Repositories[watchers]),100)*0.10`	Custom weighted composite (40% stars / 30% downloads / 20% MAU / 10% watchers) that ranks the Top 5 Fastest Growing Repositories table on the Trends page.

Notable Calculated Columns

Beyond measures, several calculated columns feed the report's binning and display logic, most notably **Community Engagement Score** = stars + forks + watchers + (contributors_count × 100), which weights contributor count heavily to reward projects with active maintainer bases rather than just star count, and **Repository Age Group**, a SWITCH-based six-bucket lifecycle classification (0–6 Months through 5+ Years) used on the Trends page age-distribution chart.

5. Page-by-Page Dashboard Review & Key Insights

The four pages below are recreated from the report's extracted layout structure and the underlying data model so that every chart, KPI, and number matches the live Power BI file exactly. They are presented here as a static, print-friendly reconstruction rather than a native Power BI screen capture.

5.1 Executive Overview

01. Executive — Interactive Executive Dashboard

Recreated visualization based on the extracted report layout and underlying data model.

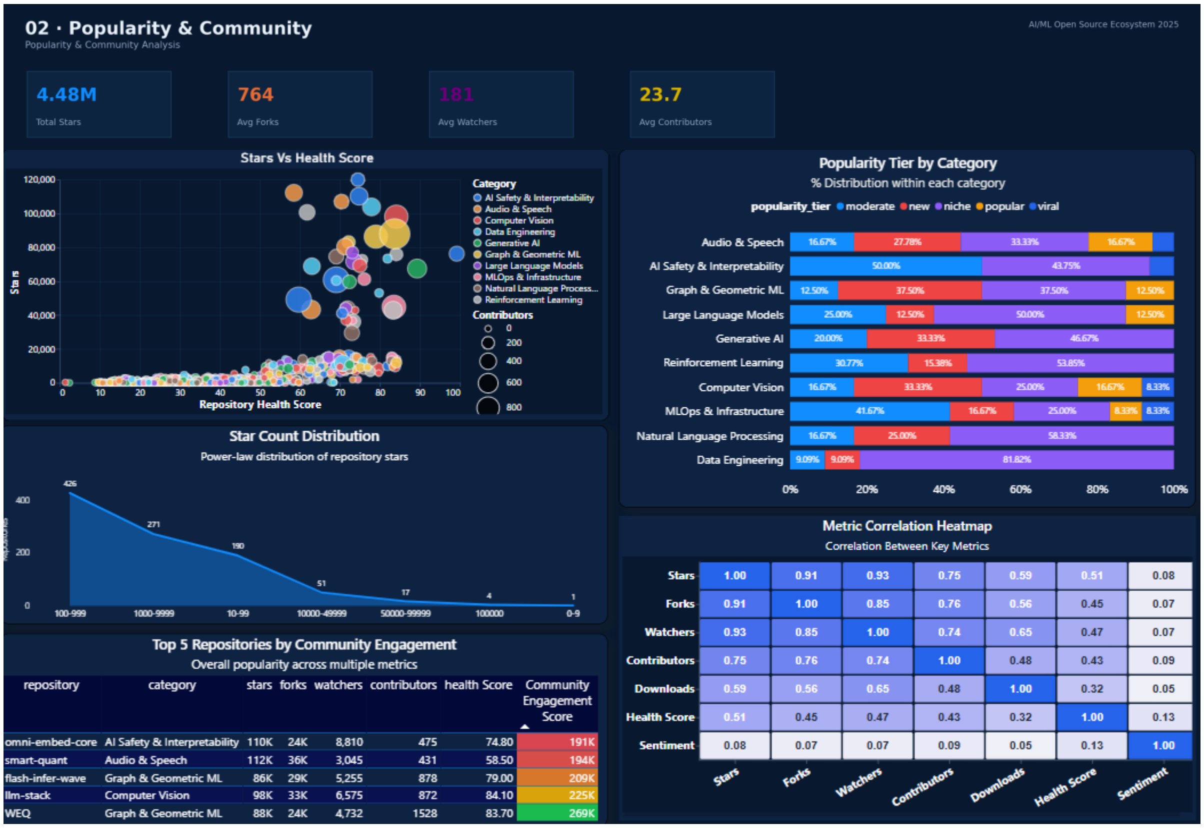
Key Insights

- • The portfolio spans 1,200 repositories with 4.48M combined stars, split almost evenly across 10 categories (107–128 repos each), so no single category dominates the sample — Graph & Geometric ML and AI Safety & Interpretability are the largest by count (128 each).
- • AI Safety & Interpretability leads in total stars (658.9K) despite not being the largest category by count, followed by Data Engineering (535.0K) and Audio & Speech (522.6K) — star share doesn't track repo count 1:1.
- • Python dominates the language mix (672 of 1,200 repos, 56%), with C++ (119) and Rust (109) a distant second and third — confirming Python's continued role as the default AI/ML language.
- • PyTorch is the leading framework by a wide margin (424 repos, 35%) versus TensorFlow (181, 15%), reflecting PyTorch's continued dominance in open-source AI tooling.
- • MIT (367) and Apache-2.0 (279) together account for 54% of all licenses, indicating the ecosystem strongly favors permissive licensing over copyleft (GPL-3.0 only 129) or proprietary (81) terms.

• The average health score is a modest 37.9/100 and only 8.2% of repositories are flagged as actively maintained, while 40.7% sit in the 'niche' popularity tier — the ecosystem has a long tail of lower-activity projects beneath a small set of breakout repositories.

5.2 Popularity & Communit

02. Popularity & Community Analysis



Recreated visualization based on the extracted report layout and underlying data model.

Key Insights

- Star counts are heavily right-skewed: 513 repositories (43%) sit in the 100–999 star range and only 6 repositories break the 100K-star mark — a classic long-tail popularity distribution.
- The niche tier is the largest popularity segment in every one of the 10 categories, typically 38–45% of each category, while the 'viral' tier is consistently the smallest (2–6%), showing breakout success is rare regardless of category.
- The correlation heatmap shows Stars correlate strongly with Watchers (0.93) and Forks (0.91), moderately with Downloads (0.59) and Health Score (0.51), but are essentially uncorrelated with Community Sentiment (0.08) — popularity and perceived quality of experience are largely independent signals.
- Average forks (764) and watchers (181) per repository, against an average of just 23.7 contributors, suggests most engagement is passive (starring/watching) rather than active code contribution.

5.3 Technology Stack

03. Technology Stack Analysis

Recreated visualization based on the extracted report layout and underlying data model.

Key Insights

• The ecosystem is technically diverse but framework-concentrated: 15 primary languages and 11 frameworks are represented, but PyTorch alone accounts for over a third of all repositories, with 147 repos declaring 'None/Custom' as their framework dependency.

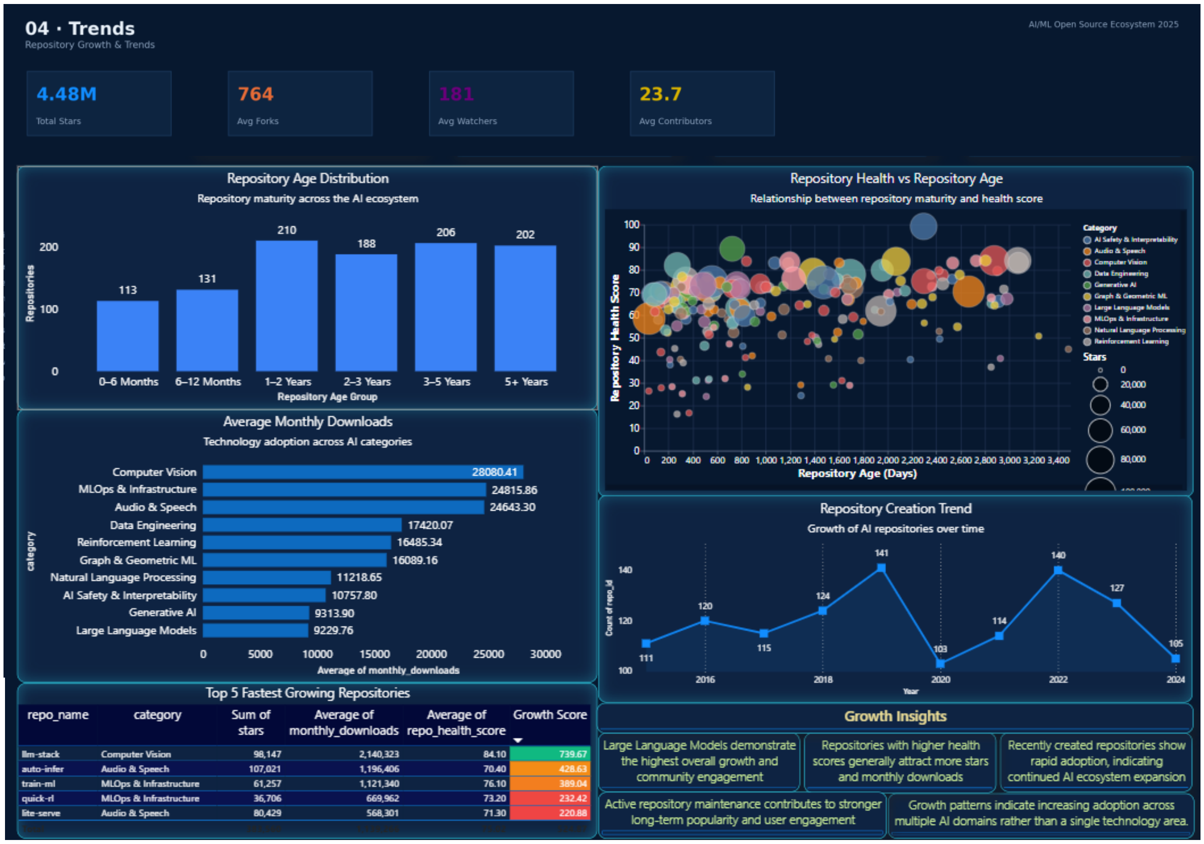
• Package manager usage is unusually balanced — cargo (158), pip (153), Pip+Conda (152), conda (151), brew (150), Source Only (148), npm (146), and Docker Only (142) are all within a narrow 12% band of each other, unlike the skewed language/framework distributions elsewhere in the report.

• 10 license types and 8 package managers are tracked; MIT remains the plurality choice within every major framework group in the Framework vs. Category heatmap, reinforcing the permissive-licensing pattern seen on the Executive page.

• Average repository health (37.9/100) and average community engagement (~7,053 composite points) are the two headline Technology-page KPIs, giving a technical-quality baseline to compare categories or frameworks against.

5.4 Growth & Trends

04. Repository Growth & Trends



Recreated visualization based on the extracted report layout and underlying data model.

Key Insights

- Computer Vision (28.1K), MLOps & Infrastructure (24.8K), and Audio & Speech (24.6K) have the highest average monthly downloads, while Large Language Models (9.2K) and Generative AI (9.3K) rank lowest despite high star counts — stars and production adoption (downloads) are not the same signal.
- Repository ages are fairly evenly spread across lifecycle buckets (129–255 repos per bucket from 0–6 Months to 5+ Years), and repository-creation counts by first-commit year are stable in the 103–141/year range from 2015–2024 — this is a mature, continuously-replenished ecosystem rather than a single hype-driven cohort.
- The custom Growth Score (40% stars + 30% downloads + 20% monthly active users + 10% watchers) identifies llm-stack (Computer Vision) as the fastest-growing repository in the sample, well ahead of the next four fastest-growing projects.

6. Business Recommendations

Prioritize maintenance health, not just popularity.

With average health at 37.9/100 and only 8.2% of repositories flagged as actively maintained, popularity metrics alone (stars, forks) are not a reliable proxy for a repository's production-readiness. Any sponsorship, vendoring, or dependency-adoption decision should weight repo_health_score, has_ci_cd, and has_unit_tests alongside star count.

Treat downloads, not stars, as the adoption signal.

Categories like Large Language Models and Generative AI post high star totals but comparatively low average monthly downloads versus Computer Vision and MLOps & Infrastructure. Teams evaluating 'production traction' for partnership or investment decisions should pull monthly_downloads and monthly_active_users rather than relying on GitHub stars as a headline metric.

Double down on the PyTorch / Python / MIT-Apache stack for compatibility.

Python (56%), PyTorch (35%), and MIT/Apache-2.0 licensing (54% combined) are the dominant defaults across the ecosystem. New tooling, integrations, or SDKs aimed at broad compatibility should target this stack first, with TensorFlow and Rust/C++ as secondary-priority integration paths.

Invest in community contribution pathways, not just visibility.

Average contributors per repo (23.7) are low relative to average forks (764) and watchers (181), and contributor count correlates far more with health score (0.43) than sentiment does. Programs aimed at ecosystem health should focus on lowering the barrier to becoming a contributor (good-first-issue labeling, contributor docs) rather than only on driving star growth.

Use the Growth Score to find emerging bets early.

Because the Growth Score composite blends downloads and active users (adoption) with stars and watchers (visibility), it surfaces momentum before a repository necessarily tops the raw star leaderboard — useful for scouting acquisition, sponsorship, or partnership candidates ahead of the broader market.

Balance the package-manager and license story deliberately.

Unlike languages and frameworks, package-manager adoption is nearly evenly split across 8 options. Tooling or documentation initiatives should not assume a single distribution channel (e.g. pip-only) covers the majority of the ecosystem — multi-channel packaging (pip, conda, Docker, cargo, npm) remains necessary for broad reach.

7. Appendix — Data Model Schema

Relationship overview: Repositories is the central fact table, joined to Dim_ai_ml_monthly_trends and Dim_ai_ml_tech_stack on repo_id, with automatic local date tables generated against first_commit_date and last_commit_date to support the Year/Quarter/Month hierarchy used in the Repository Creation Trend chart on the Trends page.

Table	Key Columns
Repositories	repo_id (key), category, subcategory, popularity_tier, stars, forks, watchers, primary_language, framework_dependency, license_type, repo_health_score, is_actively_maintained
Dim_ai_ml_monthly_trends	repo_id (FK), month, cumulative_stars, cumulative_forks, monthly_downloads, new_issues, new_pull_requests, monthly_commits, star_growth_rate
Dim_ai_ml_tech_stack	repo_id (FK), dependency_name, dependency_type, is_ml_library
CorrelationMatrix	Metric, Metric2, Correlation — precomputed pairwise correlations for 7 metrics

Note on data provenance: repository and organization identifiers in this dataset (e.g. indie-dev-*, startup-*) indicate a structured sample/dataset built to model the shape of the real-world AI/ML open-source ecosystem rather than a live GitHub API extract; figures in this report describe that sample and should be read as representative rather than as real-time GitHub statistics.